

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA0612	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Kaviya R	Reg. No.:	192521075
Section / Batch:	B tech IT	Date:	22-07-2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q53. Application: Medical Diagnosis Support System

A medical diagnosis system analyzes patient data using the recurrence $T(n) = 2T(n/4) + n$. Apply the Master Theorem to determine the time complexity. Identify parameters and justify the case selection. Analyze how this affects performance for large hospital datasets. Interpret efficiency using asymptotic notation.

Q54. Application: Satellite Image Analysis System

A satellite image analysis system processes image regions using the recurrence $T(n) = 7T(n/2) + n^2$. Apply the Master Theorem to determine the time complexity. Clearly identify parameters and justify the selected case. Analyze how processing time grows with image resolution. Interpret efficiency using asymptotic notation

Code of Conduct

I Kaviya(192521075) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Kaviya R

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____

Date: _____

Date: _____

Date: _____

CSA0612-Design And Analysis Of Algorithms

CO1 Application Based Question

Q53. Application: Medical Diagnosis Support System

Recurrence: $T(n) = 2T(n/4) + n$

Step 1: Identify Parameters

Parameter	Value	Meaning
a	2	Number of subproblems the data is divided into
b	4	Factor by which patient-data size shrinks per subproblem
f(n)	n	Cost of dividing data and combining diagnostic results

Step 2: Compute $n \log_b a$

$\log_a b = \log_2 4 = 2$, so $n^{\log_b a} = n^2 = \sqrt{n} \cdot n$

Step 3: Compare f(n) with $n \log_b a$

$f(n) = n$ grows strictly faster than n^2 . In fact $f(n) = \Omega(n^{2+\epsilon})$ for $\epsilon = 0.5$, since $n = n^{0.5} \cdot n^{0.5}$, which polynomially dominates n^2 . This places the recurrence in **Case 3** of the Master Theorem.

Step 4: Verify the Regularity Condition

We need $a \cdot f(n/b) \leq c \cdot f(n)$ for some $c < 1$. Here: $2 \cdot f(n/4) = 2 \cdot (n/4) = n/2 = 0.5 \cdot f(n)$. Taking $c = 0.5 < 1$ satisfies the condition, so Case 3 applies validly.

Step 5: Conclusion

$T(n) = \Theta(n)$ — the system runs in linear time.

Performance Analysis for Large Hospital Datasets

Because the recurrence resolves to $\Theta(n)$, the diagnosis system's processing time scales directly and proportionally with the number of patient records n . Doubling the dataset size roughly doubles the processing time, with no extra logarithmic penalty. This linear growth is highly desirable in a hospital setting: as electronic health records

accumulate and patient volume grows, the system remains predictable and scalable. The dividing cost $f(n) = n$ dominates the recursive work (2 subproblems of size $n/4$ contribute comparatively little), so the overall efficiency is governed almost entirely by the linear combination/aggregation step rather than by the depth of recursion.

Asymptotic Interpretation

$T(n) = \Theta(n)$ means the algorithm is asymptotically tight-bounded above and below by a linear function of n . This is an efficient, scalable complexity class — comparable to a single linear scan of the data — making the system suitable for real-time or near-real-time diagnostic support even as hospital data volumes grow into the millions of records.

Q54. Application: Satellite Image Analysis System

Recurrence: $T(n) = 7T(n/2) + n^2$

Step 1: Identify Parameters

Parameter	Value	Meaning
a	7	Number of sub-regions each image is split into
b	2	Factor by which image resolution/side-length shrinks
f(n)	n^2	Cost of combining sub-region results (proportional to image area)

Step 2: Compute $n \log_b a$

$\log a = \log 7 \approx 2.807$, so $n^{\log_b a} = n^{2.807}$ (this is the same recurrence structure as Strassen's matrix multiplication algorithm).

Step 3: Compare f(n) with $n \log_b a$

$f(n) = n^2 = n^{2.0}$, while $n^{\log_b a} = n^{2.807}$. Since $2.0 < 2.807$, $f(n) = O(n^{2.807 - \epsilon})$ for $\epsilon = 0.807$ (or any ϵ up to ~ 0.807), meaning $f(n)$ grows polynomially slower than $n^{\log_b a}$. This places the recurrence in **Case 1** of the Master Theorem.

Step 4: Conclusion

$$T(n) = \Theta(n^{\log_2 7}) \approx \Theta(n^{2.807})$$

Since Case 1 holds, the recursive branching (7 subproblems) dominates the combine cost, so the total running time is governed entirely by the recursive term, not by $f(n)$.

Growth With Image Resolution

Here n represents a linear dimension of the image (e.g., width/height in pixels), so total pixel count scales as n^2 . Because $T(n) = \Theta(n^{2.807})$, processing time grows super-linearly and even super-quadratically relative to the linear resolution n — worse than the n^2 combine cost per level alone. Practically, this means: doubling the linear resolution ($n \rightarrow 2n$) increases processing time by a factor of $2^{2.807} \approx 7\times$, not just $4\times$ (as pixel count alone would suggest) or $8\times$ (as a naive cubic approach might). For high-resolution satellite imagery (e.g., 4K, 8K, or hyperspectral scans), this factor-of- ~ 7 growth per resolution doubling becomes a significant computational bottleneck, and highlights why reducing the branching factor a (e.g., algorithmic optimizations akin to Strassen's algorithm reducing 8 recursive multiplications to 7) is valuable at scale.

Asymptotic Interpretation

$T(n) = \Theta(n^{2.807})$ is a polynomial but super-quadratic complexity class. It is more efficient than a naive $\Theta(n^3)$ approach (e.g., 8-way recursive splitting) but noticeably less efficient than linear or quadratic

algorithms. For satellite image analysis systems processing increasingly large or numerous high-resolution images, this complexity signals the need for careful resource planning, parallelization, or algorithmic refinement to keep processing times manageable as n grows.

Summary Comparison

	Q1: Medical Diagnosis	Q54: Satellite Image
Recurrence	$T(n) = 2T(n/4) + n$	$T(n) = 7T(n/2) + n^2$
a, b	$a=2, b=4$	$a=7, b=2$
$n^{(\log_b a)}$	$n^{0.5}$	$n^{2.807}$
$f(n)$	n	n^2
Case	Case 3	Case 1
Result	$\Theta(n)$	$\Theta(n^{2.807})$